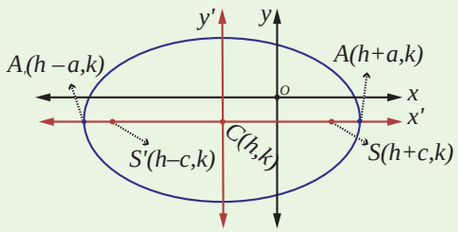
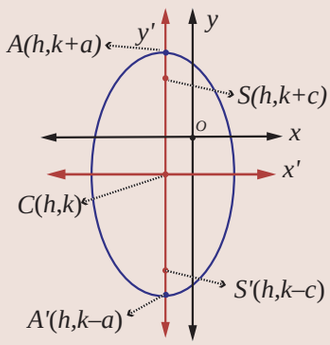
Equation	Centre	Major Axis	Vertices	Foci
$\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1 \quad \boxed{a^2 > b^2}$  Fig.5.24 (a) Major axis parallel to the x-axis Foci are c units right and c units left of centre, where $c^2 = a^2 - b^2$.	(h,k)	parallel to the x-axis	$(h-a,k)$ $(h+a,k)$	$(h-c,k)$ $(h+c,k)$
$\frac{(x-h)^2}{b^2} + \frac{(y-k)^2}{a^2} = 1 \quad \boxed{a^2 > b^2}$  Fig.5.25 (b) Major axis parallel to the y-axis Foci are c units right and c units left of centre, where $c^2 = a^2 - b^2$.	(h,k)	parallel to the y-axis	$(h,k-a)$ $(h,k+a)$	$(h,k-c)$ $(h,k+c)$

Theorem 5.5

The sum of the focal distances of any point on the ellipse is equal to length of the major axis.

Proof

Let $P(x,y)$ be a point on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

Draw MM' through P, perpendicular to directrices l and l'.

Draw $PN \perp$ to x-axis.

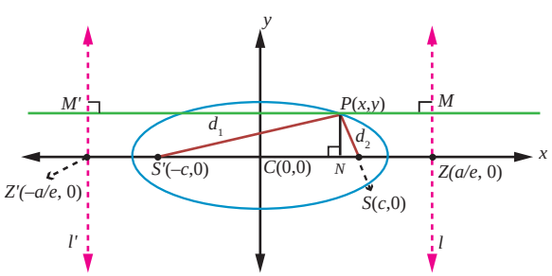


Fig.5.26

By definition

$$SP = ePM$$
$$= eNZ$$
$$= e[CZ - CN]$$

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